raviadityanarayan@gmail.com
adityanarayan.web.illinois.edu/
Adityanarayan96
narayan-aditya

Aditya Narayan Ravi

Education

University of Illinois at Urbana Champaign Ph.D., Electrical & Computer Engineering *2021 – Present*
Indian Institute of Technology Bombay B.Tech & M.Tech, Electrical Engineering *2015 – 2020*

Select Publications

Selected from a total of 14 publications:
Non-Adaptive Group Testing With Markovian Correlation, *Submitted to AISTATS 2025**
Scalable Batch Correction for Cell Painting via Batch-Dependent Kernels and Adaptive Sampling, *Submitted to NeurIPS 2025**
Prioritized Federated Learning: Leveraging Non-Priority Clients for Targeted Model Improvement, *Transactions in Machine Learning Research (jmlr.org/tmlr)**
LexicHash: Sequence Similarity Estimation via Lexicographic Comparison of Hashes, *Bioinformatics*†
* First author, † Co-author

Selected Applied Research Experience

- **Batch Correction in Cell Painting** *Aug 2024 – Present*
  - Developed and released BALANS and cPCA-cellpainting, well-documented repositories for **batch correction**.
  - Achieved over **35%** overall improvement on **deep learning** features extracted from **cell painting** datasets.
  - Achieved a **state-of-the-art runtime**, outperforming methods by over **3 hrs** on large scale data (5M data points).
- **Client Priority in Federated Learning** *Jan 2023 – July 2023*
  - Modelled a priority-based setup and algorithm (FedALIGN) in **federated learning**, improving **FedAvg** by 20%.
  - Implemented **CNNs** and **RNNs** to further test model efficiency on complex datasets.
- **Spatial Gene Expression Estimation** *Feb 2024 – Present*
  - Designed a **boolean compressive sensing** test scheme using **single cell** derived **gene-regulatory** networks.
- **Sequence Similarity Estimation: LexicHash** *Jan 2022 – July 2023*
  - Introduced a novel **similarity estimation** method, improving accuracy by 20.9% over baseline **MinHash**.
  - Developed an efficient algorithm (implemented on Python's **FASTX-toolkit**) for top-k alignment.
- **Unreliable Multi-Armed Bandits** *Aug 2019 – Jan 2020*
  - Proposed a variant of **Multi-Armed Bandits** for **recommendation systems**, modeling unreliable bandits.
  - Designed a perturbation-based algorithm, reducing convergence time by up to 33% and regret by up to 25%.

Work Experience

- **AbbVie** - **Machine Learning Research Intern** *Aug 2024 – Aug 2025*
  - Adapted and implemented **deep learning-based methods** for the **computer vision research team**.
  - Deployed high-performing models to **production on AWS**, enabling scalable **API integrations**.
  - Utilized **version control** and collaborative workflows (branching, pull requests, CI/CD) to ensure **reproducibility**.

Information Theory and Applied Mathematics Research

- 8 first-author papers [scholar] at flagship **information theory** conferences and journals (**ISIT**, **JSAIT**).
- Solved open problems in out-of-order channels and proved **theoretical results** for Gaussian Broadcast channels.

Relevant Skills

Programming Languages: Python, R, C++, C **Packages**: PyTorch, Keras, Tensorflow, Pandas, SciPy, Numpy

Academic Achievements

- **Shun Lien Chuang Memorial Award** for Excellence in Graduate Education (Awarded to 1 student)
- Twice awarded the **Joan and Lalit Bahl Fellowship** for Outstanding Graduate Studies (Awarded to 2 students)
- **IITB Undergraduate Research Award - III** for exceptional thesis (Awarded to 10/1097)